Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. **Team name**: Viskotu
b. **Team leader name**: L S Thoshi Babu
c. **Problem Statement**: Medical research already holds proven routines for injury recovery, sleep and strength. But they’re buried in dense papers written for researchers, so people guess, push through pain, or give up. The knowledge never reaches them.
d. **Team** : M.F.M.Fazarath , L S Thoshi Babu

ViskotuN1 turns dense medical research into a daily routine you can actually follow.

An AI companion built on Gemini, so you never have to read the paper.

WHY IT'S DIFFERENT

Do your exercises 12-day streak
Same plan for everyone
Pain 7/10 "Don't break your streak!"

Habit apps

One checklist for everyone, and streaks that push you on even when you're in pain.

Try 3 × 20 reps daily not in the paper

AI chatbots

A wall of text, and a made-up dose when the paper never gave one.

Heel drops 3 × 15 ✓ from the paper
Rest 60 s default, labelled
Pain 5/10 paused · 20% lighter

ViskotuN1

Built from the study: human trials only, gaps labelled, and it holds you back when pain crosses the limit.

HOW IT WORKS

~10 s

1 Drop in a study

PDF or DOI. Gemini turns the Methods into the exact sets, reps, holds and timing. Gaps are flagged, never guessed.

2 The screen fits the study

Pain sliders for rehab, hold timers for stretches, progress photos for visible change. Never just a checkbox.

4/10 → -20%

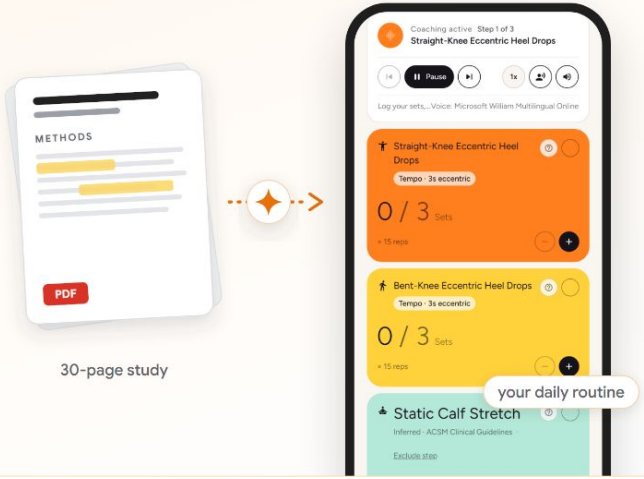
3 Coach and safety net

A voice coach guides every set, and answers come from the paper. Pain past 4/10? Tomorrow's load drops 20%.

THE PROMISE

Proven research, turned into your daily routine in about 10 seconds.

- Every number comes from the paper, or is clearly marked.
- Your body sets the pace.



Process flow diagram

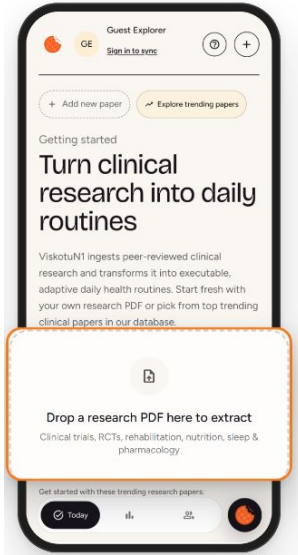


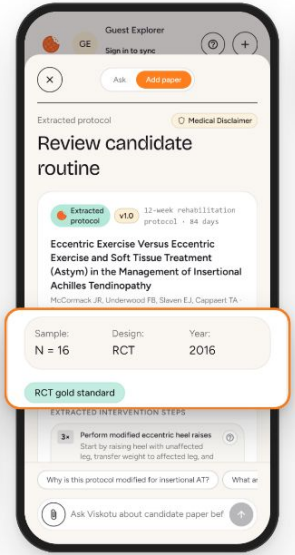
ALWAYS ON

- **Ask anything** — answered from the paper, not the internet
- **Clinical team in control** — sets the pain limit, approves AI trackers
- **Shared knowledge** — confirmed routines, reviews, versions
- **Guest or signed in** — start instantly, sync across devices

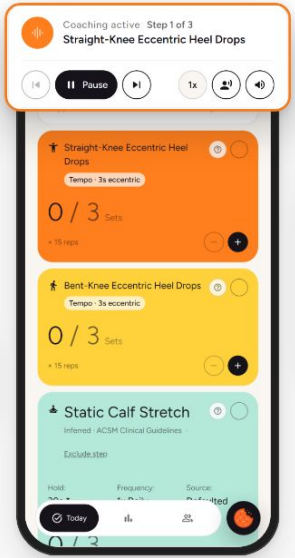
Mock diagrams of the proposed solution

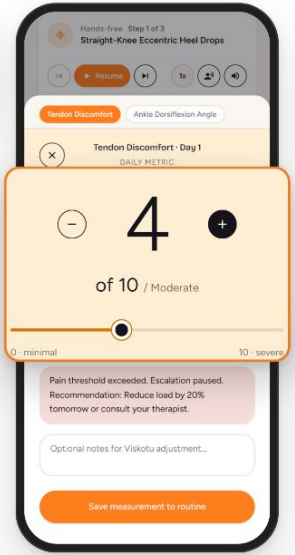
SET UP ONCE

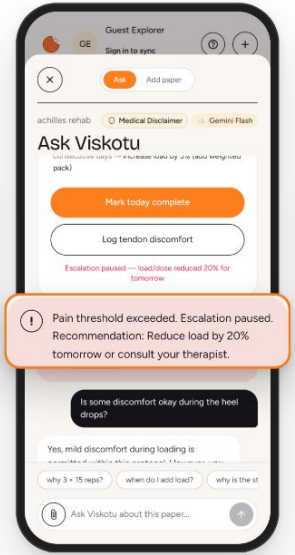




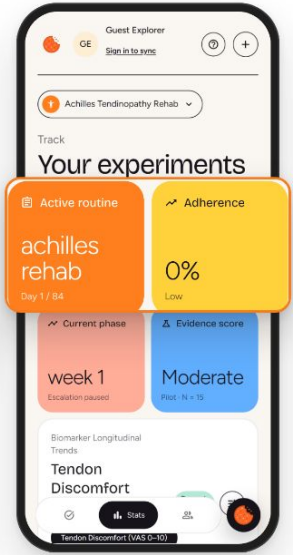
EVERY DAY







OVER THE WEEKS



1 Add a study

Drop in a PDF, paste a DOI, or pick a trending trial.

2 Check it first

See the sample, design and exact steps before you commit.

3 Follow along

The voice coach counts every hold and set, hands-free.

4 Say how it feels

Trackers come from what the study measured.

5 Safety steps in

Over the pain limit? Tomorrow gets 20% lighter.

6 See it working

Adherence, phase and trends, plus a brief for your doctor.

Real screens from the live app on Google Cloud Run. Example study: McCormack et al., Sports Health (2016), an Achilles tendinopathy trial.

List of features offered by the solution

WHEN YOU ADD A PAPER

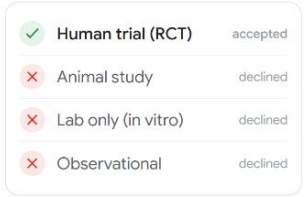
Paper to Routine

Drop a PDF or DOI and get an interactive daily routine in seconds, ready for you to review.



Science Gatekeeper

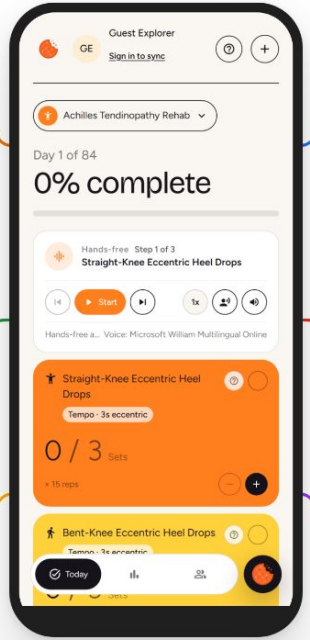
Accepts only human intervention studies and rejects animal, lab-only, theoretical and observational papers, with the reason.



Missing-Data Flags

Shows what the paper left out, labels every conservative default it fills in, and lets you exclude it in one tap.

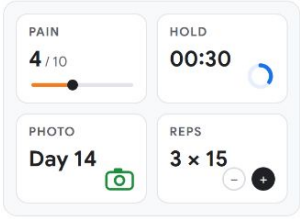




EVERY DAY

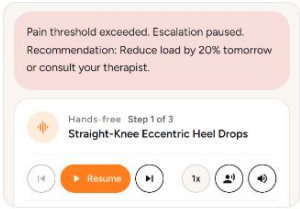
Adaptive Tracking

Sliders, timers, photo milestones or number inputs, matched to what the study measured.



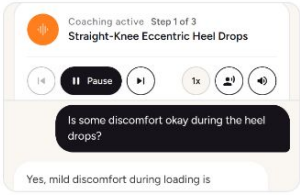
Pain Auto-Pause

Holds your progression and suggests a 20% lighter load when pain reaches the clinician-set safety limit (default 4/10).

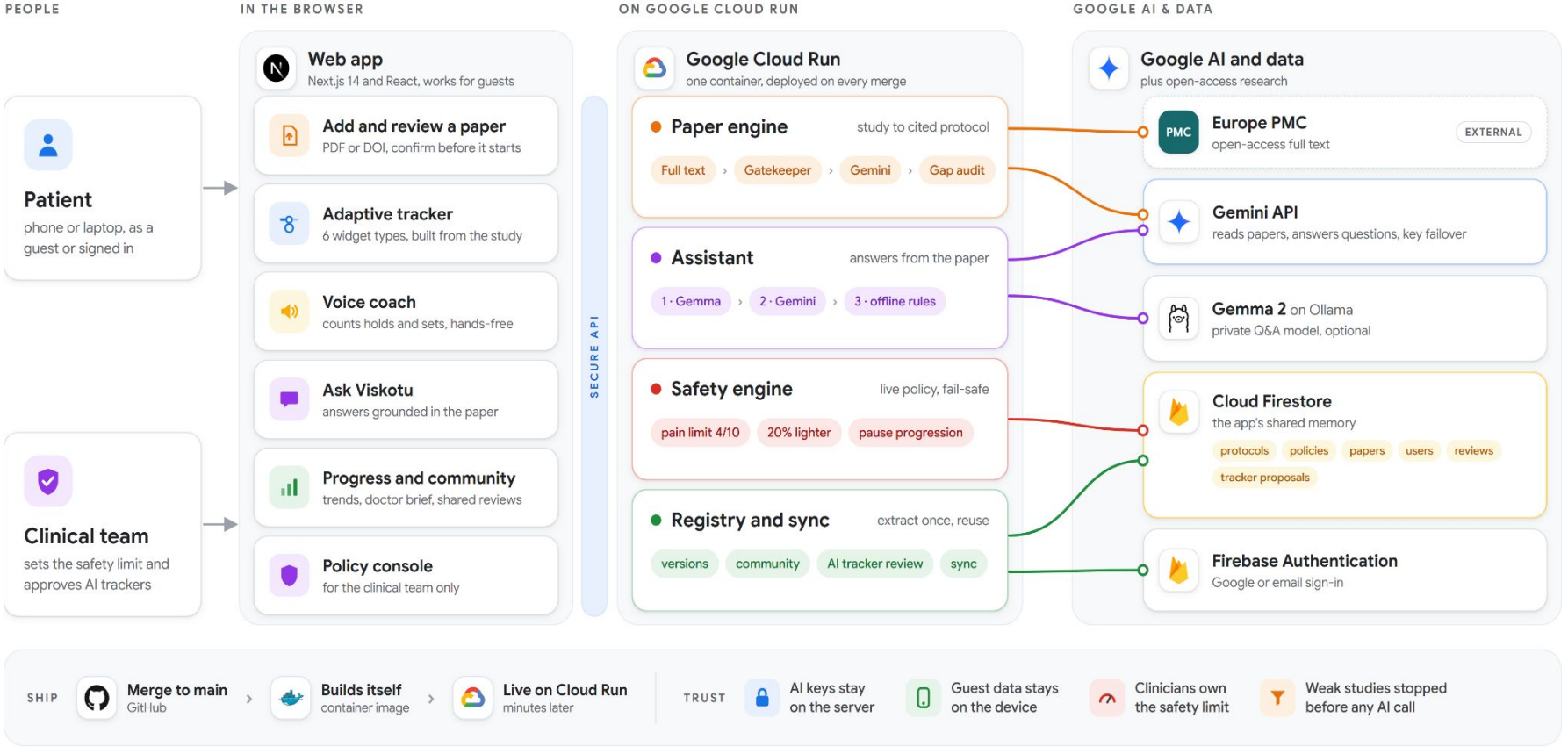


Voice Coach & Paper Chat

Guides your sets out loud and answers questions grounded in the paper.



Architecture diagram of the proposed solution



Tech Stack

WHAT PEOPLE USE

A fast web app

phone or laptop, and it works for guests with no account

- **Next.js 14** — pages and API
- **React 18** — interface
- **TypeScript** — end to end
- **Tailwind CSS** — styling
- **Framer Motion** — motion
- **Material icons** — icons
- **Recharts** — trend charts
- **Web Speech** — coach's voice

THE INTELLIGENCE

AI that reads the paper

so the patient never has to

- **Gemini API** (Google) — Reads the Methods and writes a cited, step-by-step protocol
- **Gemma 2 on Ollama** (Google) — Answers daily questions on a private server
- **Europe PMC** — Open-access full text from a DOI or PubMed ID
- **Cloud Text-to-Speech** (Google) — A natural voice for the coach

THE FOUNDATION

Google Cloud underneath

hosting, data and sign-in

- **Cloud Run** — Runs the whole app in one container. Every push to main goes live.
- **Cloud Firestore** — Protocols, safety limits, reviews and synced progress
- **Firebase Authentication** — Google sign-in when you want it. Guests need no account.

HOW WE SHIP

Tested, then live

every change is reviewed first

- **GitHub** — merge to main, live in minutes
- **Docker** — one standalone image
- **Node.js** — server runtime
- **Vitest** — 229 tests, all passing
- **ESLint** — zero warnings

Logos belong to their owners and are shown only to name the technology used.

Estimated implementation cost

PROTOTYPE TODAY
≈ $0
runs inside the Google Cloud and Gemini free tiers

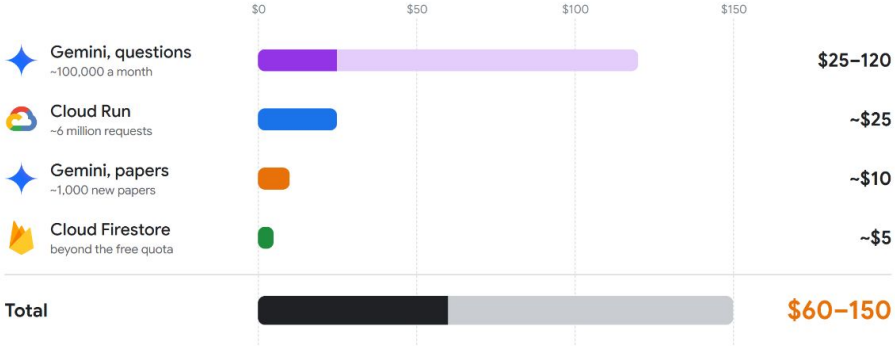
PER PAPER
< 1¢
paid once, then the protocol is reused by everyone

PER QUESTION
~0.1¢
only when Gemma can't answer and Gemini steps in

PER USER
1–2¢ / month
at about 10,000 monthly active users

What 10,000 users would cost each month

Cloud Run, Firestore and Gemini together · darker bar is the low estimate, lighter is the high one



Why it stays cheap

Read once, reuse forever
Gemini reads each paper one time. Every later user gets the confirmed protocol from the registry for free.

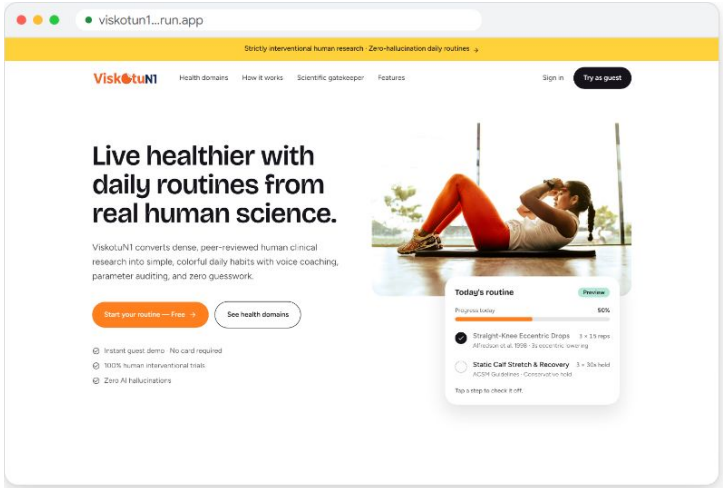
Gemma answers first
Daily questions go to Gemma on a private server. Gemini is only the fallback.

Pay only while serving
Cloud Run scales to zero when idle, and Firestore's free quota covers prototype traffic.

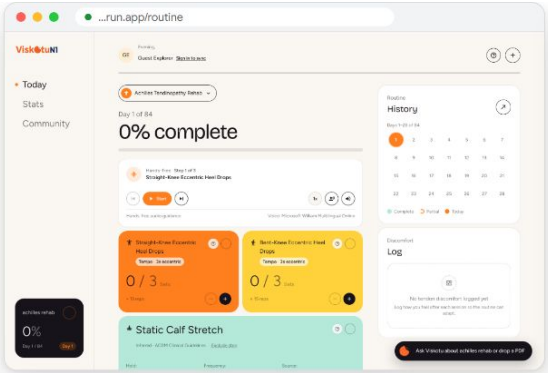
+ **Hosting Gemma:** a small VM at about $50–100 a month, or a Cloud Run GPU for faster answers.

Gemini 3.1 Flash-Lite list price: $0.25 per million input tokens, $1.50 per million output. Usage figures are estimates.

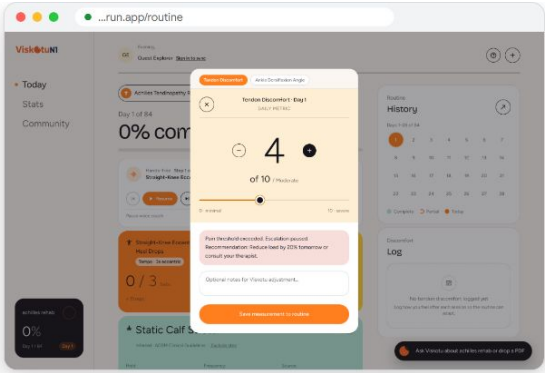
Snapshots of the prototype



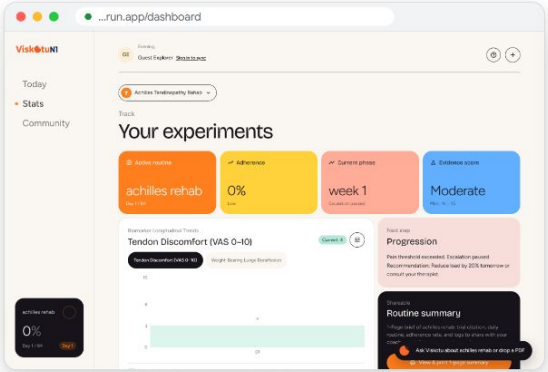
1 First look one clear promise, try it as a guest



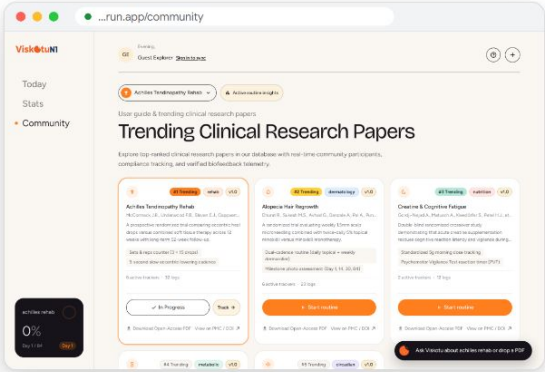
2 Today voice coach, set counters, history



3 How it feels pain 0–10, safety rule built in



4 Progress adherence, trends, 1-page brief



5 Community trending trials with real PMC links

LIVE ON GOOGLE CLOUD RUN

Scan to try it

`viskotun1-mono-git-536575310594.europe-west1.run.app`

Guest mode starts instantly.
No sign-up, nothing to install.

Benchmarking

PAPER TO PROTOCOL
8–11 s
from a DOI link to a cited, step-by-step protocol

TURNED AWAY
~1 s
to reject an unsuitable study, with no AI call spent

SAFETY CHECK
10–60 ms
to read the clinician's pain limit

VOICE COACH
~20 ms
for each request to the voice service

PAGES
0.1–0.9 s
server response for any page on Cloud Run

Does it match the paper?

Achilles tendinopathy trial · McCormack et al., Sports Health (2016)

	THE PAPER SAYS	VISKOTUN1 EXTRACTED	
Participants	16 people	**N = 16**	✓
Dose	3 sets of 15 reps	**3 × 15**	✓
How often	twice a day	**2 × daily**	✓
How long	12 weeks	**84 days**	✓

Our first version got this wrong: it only saw the link. Now it reads the full text, and every number checks out.

Quality checks

run on every change before it reaches main

229 of 229 — **Automated tests pass**
including studies the gatekeeper must turn away

0 lint warnings, and the production build is clean

12 bugs fixed in testing, each one checked in a real browser

Doing less work

safety-limit requests while a page is open

BEFORE
~30 every second, non-stop

AFTER
1–2 per page load

1× each paper is read once, then reused for everyone from the registry

60 s safety limit is cached; if the database is unreachable, a safe default applies

Single-run timings on the live Cloud Run app and in local development, 18 Oct 2026. Not a formal load test.

Future Development

NEXT
Make it trustworthy

More reliable extraction

No simulated routine when a PDF fails. Accuracy measured against hand-checked protocols.

gold-standard set · no fallback guesses

Security and compliance

Sign-in on the admin and sync APIs, audit trails, and groundwork for health-data rules.

API auth · audit trail · GDPR · HIPAA

THEN
Make it smarter and more private

Private Q&A on the phone

Connect the on-device Gemma helper, so daily questions never leave the phone.

Clinician sign-off

Clinicians approve community protocols and corrections before they become the shared version.

Smarter gap filling

Read the paper's cited references, its lineage, to fill missing details automatically.

LATER
Reach more people

Wearables

Steps, heart rate and sleep logged automatically through Health Connect.

Every language

Routines and coach voices in your own language, through Gemini and Cloud voices.

Installable app

A mobile app on the home screen, not just a website.

Beyond rehab

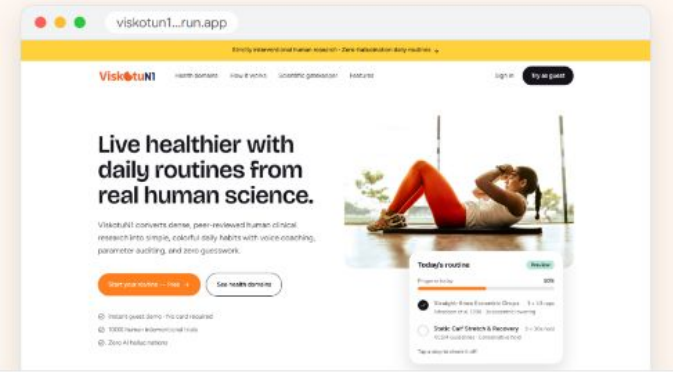
Sleep, nutrition and medication-adherence trials.

NORTH STAR
Prove what works

Measured outcomes

With consent, anonymised community data shows which published routines actually work for real people.

People log their routines
↓
Consent, then anonymised
↓
Real-world evidence for each routine

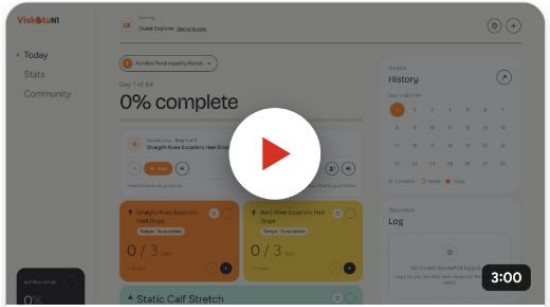


LIVE APP

Open the product

Runs on Google Cloud Run. Guest mode starts instantly, no sign-up.

Final Product Link: [Open Final Product]

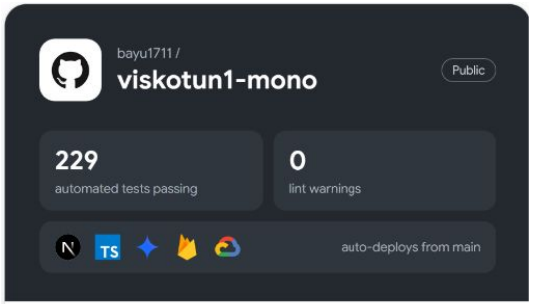


DEMO VIDEO

Watch the 3-minute demo

From a research paper to a safe daily routine, end to end, in the live app.

Demo Video Link : [Watch Demo Video]



SOURCE CODE

Read the code

Public repository: Next.js, TypeScript, Gemini and Firebase.

GitHub Repository: [GitHub Repository]

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

